Leisure Time Physical Activity (LTPA) among Filipinos and its determinants

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Background of the study

People's leisure activities are significantly different today than they were few years ago. The obvious explosion and use of communication and entertainment electronic gadgets, such as cellular phones, computers, and games consoles change the way of life if not all people but majority of them. Likewise, the socio-economic transformations of the majority countries of the world contributed significantly to the decrease in the level of physical activity of their people.

In most countries, the unprecedented social and economic changes have rapidly affected physical activity and dietary patterns, which in turn have contributed to current rise in non-communicable diseases. The burden of chronic diseases is increasing worldwide. Various studies indicate that low physical activity is thought to be an essential risk factor for developing cardiovascular diseases, overweight, obesity, diabetes, certain neoplastic growths and other chronic diseases. Regular physical activity of a recreational type is a crucial element in health promotion, maintenance of good heath status and prevention of chronic diseases.

Taking into account the changes on the way people live nowadays, it is important to know and learn how the Filipino use their leisure time. What are the factors affecting their activities during their leisure time? Everybody would agree the importance of being active in physical activities especially during their leisure time in order to become healthy. However, the challenge of internet, social media, and other electronic communication devices greatly affected the time available for physical activities.

It has been estimated that chronic diseases contributed to about 60% of the total 56.5 million deaths reported throughout the world and to about 46% of the global burden of disease. The proportion of the burden of non-communicable

diseases is expected to increase to 57% by 2020.[1]

Review of related literature

On the paper of Aguiar, M. and Hurst, E. (2005) titled *"Measuring Trends in Leisure: Evidence from Five Decades of Time Use Surveys"* found out that there was a dramatic increase in leisure time lies behind the relatively stable number of market hours worked (per working age adult) between 1965 and 2003. They documented that the increased in leisure is made possible by the very large decline in time spent in non-market production. Further, they found out that leisure increased during the last forty years for a number of sub samples of the population, including samples defined by employment status, martial status, or the level of educational attainment. However, those with a high school education or less experienced the largest increases in leisure time, complementing the fact that their market work hours decreased more (males) or increased less (females) than their more educated counterparts. This divergence in leisure between the high and low educated started to become pronounced during the 1980s. They concluded that a growing "inequality" in leisure that is the mirror image of the growing inequality of wages and expenditures, making welfare calculation based solely on the latter series incomplete.

Participation in leisure-time physical activity appears to follow a socioeconomic gradient. Martínez-González, MA., et al (2001) and Borodulin K., et.al (2008) found out that low education are associated with a low level of physical activities during leisure time. Paper of Burton NW and Turrell G. (2000) titled *"Occupation, hours worked, and leisure- time physical activity"* found out that occupational class likewise associated with a low level of physical activities during leisure time.

Several authors (Burton NW and Turrell G., 2000), (Salmon, J., et.al, 2000) and Popham, F., et.al, 2006) found out that people in lower-status occupations or physically strenuous jobs are less likely to participate in such activity. In contrast, authors Wu, B. & Porell, F., (2000) and Wolin, KY & Bennett, GG (2008) have

[1] World Health Organization (WHO). *The World Health Report 2002: Reducing Risks, Promoting Healthy Life.* Geneva: World Health Organization, 2002.

reported that people in more physically demanding work tend to engage in exercise more often than those in less physically demanding work.

On the papers of Lallukka, T., et.al (2004) and Kouvonen, A., et.al (2005) found out that leisure-time physical activity varies between genders. Women often have jobs that include less occupational physical activity compared to men. Generally in Europe, men are more physically active during leisure time than women. In Finland, however, the opposite is true: leisure-time and commuting physical activity are more common among women than men. It could be that, compared to men, women do more domestic physical activity, which explains their higher level of leisure-time physical activity.

Similarly, on the paper of Mäkinen, Tomi, et.al (2010), found out that inactivity during leisure time was more common in low income (blue-collar employees than in their white-collar counterparts). Occupational class describes the range of physical and psychosocial working conditions as well as time constraints that can affect an employee's willingness and physical and mental ability to engage in leisure-time physical activities. In particular, the amount of muscle work and the demands this puts on the cardiorespiratory and locomotors systems of manual workers could influence their motivation to participate in sports and exercise. Their study concluded that having a long history of exposure to physical work (among men) and a high current job strain (among women) contributed to occupational class differences in leisure-time physical inactivity.

On the paper of Lobaszewski, J., et.al (2011) titled *"Patterns of leisure time physical activity and its determinants among a sample of adults from Kielce region, Poland"* found out that a significant proportion of participants were inactive in their leisure time. Walking was the most popular form of physical activity. Urban residence, higher level of education and better financial situation were strongly associated with higher levels of physical activities. Their study further discovered that 43% of the respondents had not walked for at least 10 minutes in their leisure time during the last week. The majority did not engage in any moderate or vigorous physical activity. Being physically active was more prevalent among residents of urban areas and among respondents with a higher level of education. Performing physical activity was also positively associated with higher net income per household member. On average, respondents had spent 47 hours in a sedentary position

during last week. The majority of subjects reported watching TV for 6-20 hours per week. Not having enough time and lack of willingness were given as the main reasons for being inactive.

The paper of Peter Schnohr (2008) titled *"Physical activity in leisure time: impact on mortality"* which aims to (1) To describe the relation between physical activity in leisure time and mortality in the general population, and (2) To evaluate if the level of physical activity in leisure is related to the level of well-being, with emphasize on joggers recommended two level of physical activity in leisure time; **First step:** *From physical inactivity to light physical activity in leisure time*. The optimal would be to motivate all sedentary persons to get up from the television-armchair, and begin walking, bicycling or other light physical activities 2-4 hours a week in moderate speed. The first step according to the author would not only increase lifespan, but also well-being. Elderly persons expected to benefit even more than younger, so it is never too late to take up physical exercise. **Second step:** *Regular vigorous physical activity*. After some months with light physical activity, one should increase the physical activity to more than 4 hours a week, and further include some fast sessions of brisk walking, jogging or other vigorous physical activity several times weekly.

On the paper of Kruger, J. (2004) about the *Trends in Leisure-Time Physical Inactivity by Age, Sex, and Race/Ethnicity - United States, 1994-2004* discovered significant decline of the prevalence of leisure-time physical inactivity from 29.8% in 1994 to 23.7% in 2004. The proportions of physical inactivity among men and women were highest in 1994 (men, 27.9%; women, 31.5%) and decreased through 2004 (men, 21.4%; women, 25.9%). The largest decline was among men aged 50-59 years (from 33.5% to 23.5%) and among women aged 60-69 years (from 37.8% to 28.5%). Among racial/ethnic groups, during 1994-2004, prevalence of leisure-time physical inactivity was lowest among non-Hispanic white men and decreased from 26.4% to 18.4%. Among non-Hispanic black men, the prevalence of physical inactivity decreased from 34.2% in 1994 to 27.0% in 2004. Among Hispanic men, the prevalence of leisure-time inactivity decreased from 37.5% in 1994 to 32.5% in 2004. Non-Hispanic black women (45.7% in 1994 to 33.9% in 2004) and Hispanic women (44.8% in 1994 to 39.6% in 2004) had higher prevalence than non-Hispanic white women (28.3% in 1994 to 21.6% in 2004).

On the paper of Ahmad Taufik Jamil, et.al (2015) that aims to study the associated factors related to Non-Leisure Time Physical Activity (NLTPA) in Malaysia. The study concluded that there was a significant association between NLTPA with age ($P < 0.001$), gender ($P < 0.001$), ethnicity ($P < 0.001$), education ($P < 0.001$), occupation ($P < 0.001$), income ($P < 0.001$), marital status ($P < 0.001$), region ($P <0.001$) and residence ($P < 0.001$).

However, on the study of M. Kamil Ozer, et.al (2010) analyzing the cross-sectional data from questionnaires by 768 Turkish adults aged 40-64 years resulted that age, employment and income-expenditure rate did not affect leisure activity for both men and women in urban area, while being educated and unhealthy for women and being illiterate for men correlated with being active. In squatter area, low educated women were more active than high educated women and employed women were more active than unemployed ones, while age, income-expenditure rate, education and total health status were not related with leisure activity in squatter men. Different factors affect leisure activity in the urban and squatter areas. Physical inactivity was related with unemployment in only squatter women. There is not any relation with health status and leisure physical activity.

Statement of the problem

In recent years physical activity has become a major public health issue in most parts of the world.[2] Despite the benefits of physical activity and its important role in preventing premature deaths from cardiovascular diseases, diabetes mellitus, hypertension, obesity and certain types of cancer, 50% of adult population in the European region are estimated to be either overweight or obese and obesity related illness are estimated to account for as much as 7.0% of total health care cost due to physical inactivity.[3] According to significant number of studies, physical inactivity is a leading factor of mortality and morbidity worldwide. About 3 million deaths and 32 million disability-adjusted life years each year are due to physical inactivity.[4]

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[2] Ahmad Taufik Jamil1, Rakesh Singh1, Aniza Ismail1, Azahadi Omar2. Non-leisure time Physical Activity for adult Malaysian and determinant factors. Malaysian Journal of Public Health Medicine 2015, Vol. 15 (3): 84-93

[3] Meseguer, C. M., Galán, I., Herruzo, R., Zorrill, B. & Rodríguez-Artalejo, F. Leisure-time physical activity in a southern European mediterranean country: adherence to recommendations and determining factors. Revista española de cardiología (English ed.) 2009; 62(10):1125–33. doi:10.1016/S1885- 5857(09)73327-4

[4] Cheah, Y. K. & Poh, B. K. The Determinants of Participation in Physical Activity in Malaysia. Osong Public Health and Research Perspectives 2014; 5(1): 20-17. doi:10.1016/j.phrp.2013.12.002

In Philippines, very few studies have tried to understand the factors associated with participation in physical activity. The aim of this study is to determine the proportion of Filipino engaged in physical activities during their leisure time and what are the factors affecting it.

List of the variables to be measured

1. **Leisure Time Physical Activity (LTPA)** – is data on the subjects' leisure physical activity history that shall be collected by asking respondents whether they do exercise or play sports regularly. People who answered no were defined as being physically inactive.

Leisure is a multidimensional construct yielding many definitions and interpretations. The most straightforward definition of leisure is free or discretionary time. In other words, leisure is the time separate from a person's existence (work) or subsistence (maintaining bodily needs). Although leisure has been defined in a number of ways, including as a state of mind, as an activity, or even in a holistic fashion, it occurs when a person feels free from constraints and is motivated by internal forces.[5]

On the other hand, *Physical Activity* is defined as any bodily movement produced by skeletal muscles that result in energy expenditures.[6]

2. **Age groups -** Age was classified into three (3) categories (17 years old and below; 18 – 59 years old; 60 years old and above)

3. **Gender** – Respondent's gender (Male or Female)

4. **Education Attainment** - Respondents shall be asked to provide their highest school level completed to evaluate educational level. Responses were coded to the highest education level completed within groupings. (Less than high school; high school; some college; college or higher; and vocational)

5. **Employment -** employment status shall be determined based on whether the participant reported working at a job or business during the past year by using a standard systematic assessment of occupational involvement. Employed persons

[5] Edginton, C. R., DeGraaf, D. G., Dieser, R. B., & Edginton, S. R. (2006). *Leisure and life satisfac- tion foundation perspectives* (4th ed.). New York, NY: McGraw-Hill.

[6] Caspersen, C. J., Powell, K. E., & Christenson, G. M. (1985). Physical activity, exercise, and physical tness: De nitions and distinctions for health-related research. *Public Health Re- ports, 100*(2), 126–131.

are very heterogeneous, including soldier, lawyer, bank employee, civil servant, teacher, physician, academician, small shopkeepers, etc. The unemployed include those outside the active workforce, such as retired, homemaker, jobless. With this information, two categories of employment were established: employed and unemployed.

6. **Health condition/status** - data on the subjects' health histories shall be collected. Respondents shall provide self-assessment on the status of his/her health condition.

7. **Body Mass Index (BMI)** - The Body Mass Index (BMI) is widely used for the determination of nutritional status or body composition. It is the value obtained when the weight in kilograms is divided by the square of the height in meters. Nutritional status is determined through BMI cut-offs values as recommended by the World Health Organization (WHO), however, in recent years some scientists in the Asia-Pacific region made recommendations that the criteria values be modified to fit the Asian body composition, which is claimed to be different compared to the American or European body composition. Thus the "Asian criteria" for the determination of nutritional status based on the BMI was born and was later adopted by several Asian countries including the Philippines in 2003.[7]

8. **Income (Gross Monthly) –** for **employees**, it refers to the gross monthly wages or salaries before deduction of employee CPF contributions and personal income tax. It comprises basic wages, overtime pay, commissions, tips, other allowances and one-twelfth of annual bonuses. For **self-employed persons,** gross monthly income refers to the average monthly profits from their business, trade or profession (i.e. total receipts less business expenses incurred) before deduction of income tax. *(Note. the definitions used conform to international guidelines recommended by the International Labor Organization)*

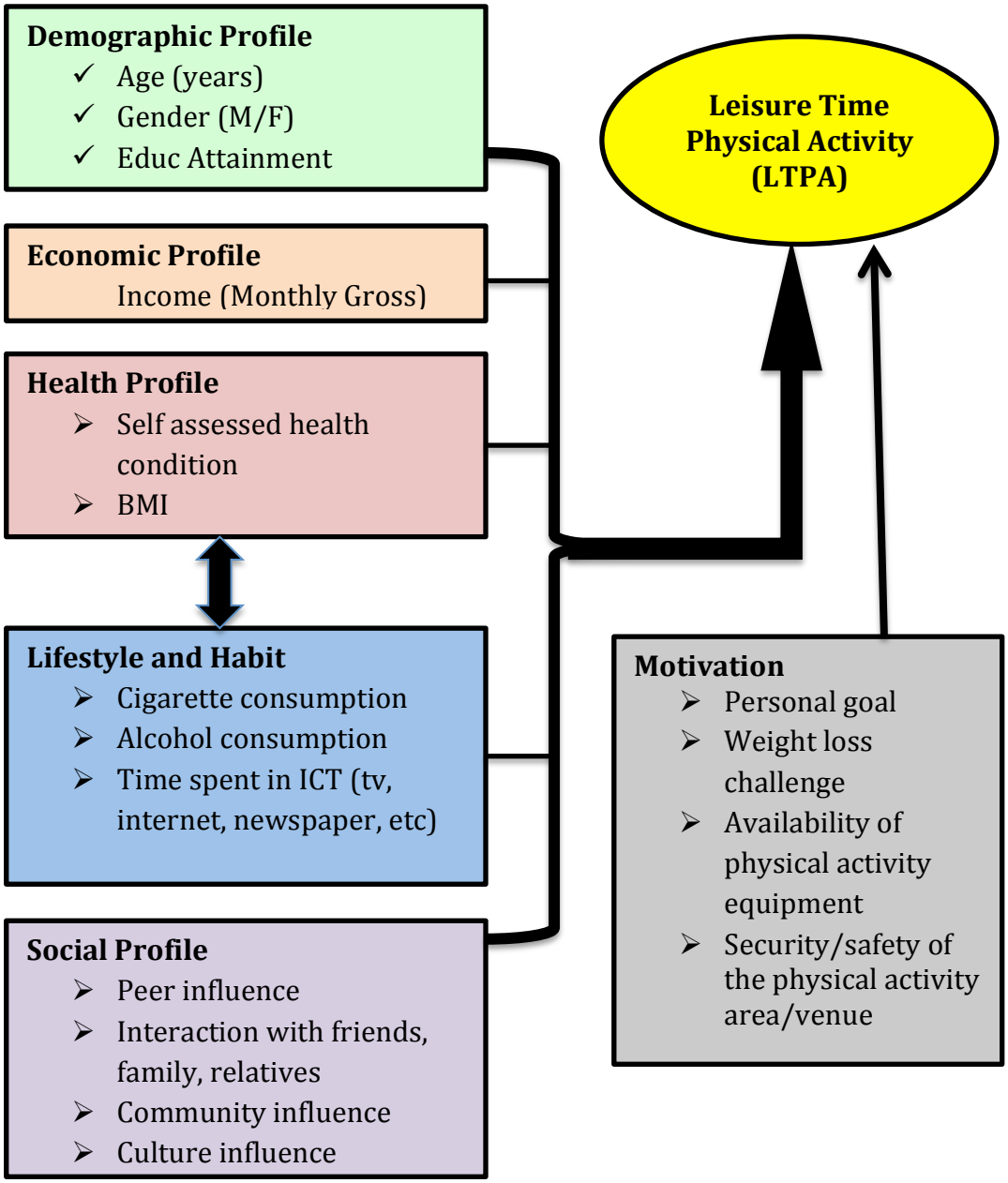
9. **Lifestyle and Habit** – Habit is defined as a set of attitudes, habits or possessions associated with a particular person or group. **Lifestyle** can be healthy or unhealthy based on your food choices, activity level and behavior.

[7] Llido, LO & Mirasol, R (2011). Comparison of Body Mass Index based nutritional status using WHO criteria versus "Asian" criteria: report from the Philippines. PhilSPEN Online Journal of Parenteral and Enteral Nutrition.

10. **Social Profile** - are a description of individuals' **social** characteristics and interaction within different people or groups. Social profile can be the level of influence the respondent received to behave in such ways.

11. **Motivation –** it is the general desire or willingness of someone to do something. Motivation can be measured through determining the degree of motivation of the respondent

Schematic diagram



Method of collecting data

The method that will be used in collecting data is personal (face-to-face) interview technique with the members of the households. The face-to-face interview is selected over the other methods of collecting data such as survey conducted through mail, telephone or online considering that it is good in gathering accurate screening responses since the individual being interviewed will be having a difficult to provide false information during the screening question. Second, face-to-face can able to capture verbal and non-verbal ques, but this method also affords the capture of non-verbal ques including body language, which can indicate a level of discomfort with the questions. Adversely, it can also indicate a level of enthusiasm for the topics being discussed in the interview. Another good for the face-to-face interview is the focus on the activity at hand. It is assumed that focus will be attained because the interviewer is the one that has control over the interview and can keep the interviewee focused and on track to completion. Hence, the interviews are in the moment, free from technological distractions. It can no doubt capture an interviewee's emotions and behaviors.

Limitations of the study

The focus of this study is the physical activities performed during free time/ leisure time such as exercise, recreation or any time other than that associated with one's regular occupation, housework or transportation.

Physical activity can be divided into Leisure time Physical activity (LTPA) and Non-leisure time physical Activity (NLTPA). LTPA is defined as physical activity performed during free time/ leisure time such as exercise, recreation or any time other than that associated with one's regular occupation, housework or transportation. NLTPA includes walking or cycling for transportation, Occupational activity is physical activity performed at work, including housework.

Statistical procedures for the study

a) Hypothesis

1. Filipino male is more likely to engage in leisure time physical activities.

2. Filipino aged 18 – 59 years old are more active in physical activities than 17 y.o and below and 60 y.o and above.

3. Social influence (social profile) greatly affects the one's leisure time activities.

4. People with higher BMI have less active physical activities during their leisure time.

b) Target population

The target population is household members who engaged (or not) in Physical Activity during their leisure time.

c) Sampling design and sample size

i) Sampling Design – Simple Random Sample (SRS) where members of the subset are chosen completely at random so that every member of the population has an equal probability of being selected

ii) Sample size determination

$$n = \frac{z^2\sigma^2}{me^2}$$

Where:

- **n** – sample size
- **z** – standard score associated 95% level of confidence (1.96)
- σ^2 – variance in data
- **me** – margin of error

d) Statistical Analysis

i) Descriptive Statistics and associated dummy tables

Table 1. Descriptive Analysis of variables

Population		Pct
1. Demographic Profile		
Age (Years)	17 years old and below	
	18 – 59 years old	
	60 years old and above	
Gender	Male	
	Female	
Education Attainment	Less than high school	
	High school	
	Some college	
	College and higher	
	Vocational	

Occupation/Employment	Employed		
	Unemployed		
Marital Status	Single		
	Married		
	Widow (er)		
2. Economic Profile			
Income (gross monthly)	=<10,000		
	10,001 – 20,000		
	20,001 – 30,000		
	30,001 – 40,000		
	40,001 – 50,000		
	50,001 – 100,000		
	>100,000		
3. Physical and Health Profile			
Body Mass Index (BMI)	*Category*	*Asian Criteria cut-off range*	
	Underweight	<18.5	
	Normal	18.5 – 22.9	
	Overweight	23 – 24.9	
	Pre Obese	25 – 29.9	
	Obese	=> 30	
	Obese Type 1	30 - 40	
	Obese Type 2	40.1 – 50	
	Obese Type 3	>50	
Self perceived health condition	Very good		
	Good		
	Fair		
	Poor		
	Very poor		
Physical disabilities	Yes		
	No		
4. Lifestyle and habit Profile			
Work Hours	<4 hrs		
	4 – 8 hrs		
	> 8 hrs		
Cigarette Consumption			
Alcohol Consumption			
Time spend for Information and communication technologies (tv, newspaper, radio, internet, etc..)			
5. Social Profile			
Interaction with family, peers, relatives, etc..			

ii) Tests of hypothesis and associated dummy tables

Table 2. Test of significance of identified variables with Leisure Time Physical Activity (LTPA)

Variable	Active leisure time physical activity	Non-Active leisure time physical activity	x^2	p-value

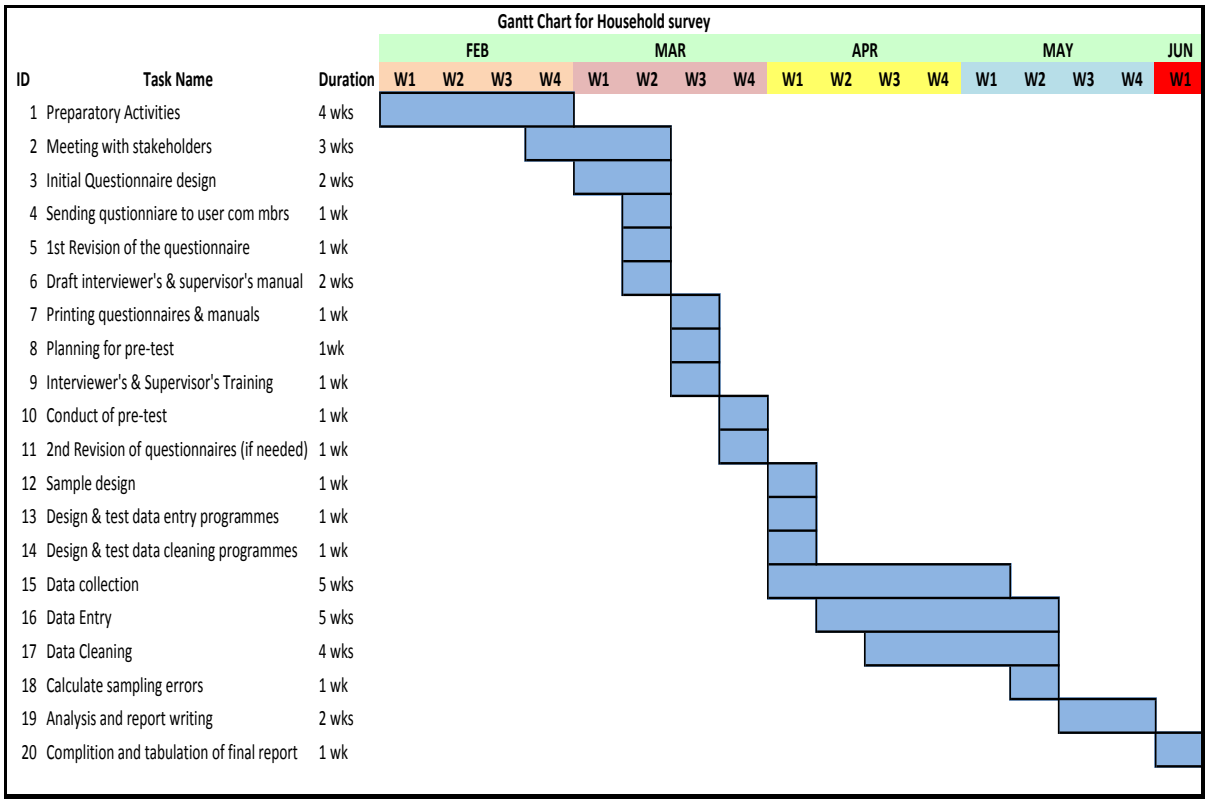
Age groups					
➢ 17 y.o and below					
➢ 18 – 59 y.o					
➢ 60 y.o and above					
Gender					
➢ Male					
➢ Female					
Educ attainment					
➢ < High school					
➢ High school					
➢ Some college					
➢ College or higher					
Employment					
➢ Employed					
➢ Unemployed					
Health status (self perceived health)					
➢ Very good					
➢ Good					
➢ Fair					
➢ Poor					
➢ Very poor					
Physical disabilities					
BMI					
Category	*Asian Criteria cut-off range*				
Underweight	<18.5				
Normal	18.5 – 22.9				
Overweight	23 – 24.9				
Pre Obese	25 – 29.9				
Obese	=> 30				
Obese Type 1	30 - 40				
Obese Type 2	40.1 – 50				
Obese Type 3	>50				
Income (gross monthly)					
➢ =<10,000					
➢ 10,001 – 20,000					
➢ 20,001 – 30,000					
➢ 30,001 – 40,000					
➢ 40,001 – 50,000					
➢ 50,001 – 100,000					
➢ >100,000					
** significance p-value <0.05*					

iii) Statistical modeling

Table 3. Multiple Logistic Regression						
Variable	β	*S.E*	*Wald*	*p-value*	*OR*	*95% CI*

OR – odds ratio
CI – Confidence interval
**significance p-value <0.05*

Gantt chart



Proposed budget

Proposed Budget					
Particulars	**Qty**	**Unit**	**Unit Price**	**Freq**	**Total**
A. Overhead cost					
Office rental (monthly)	1	room	4,500.00	4	18,000.00
Lights & Water (monthly)	1	lot	2,000.00	4	8,000.00
Office & Field Equipment	1	lot	10,000.00	1	10,000.00
Office Supplies	1	lot	5,000.00	1	5,000.00
Salary(monthly)					
Staff	2	pax	2,500.00	3	15,000.00
Interviewers	5	pax	4,500.00	3	67,500.00
Sub Total					**123,500.00**
B. Transportation & Communication					
Fare	30	pax	50.00	15	22,500.00
POL	1	lot	500.00	5	2,500.00
Cellular phone load	3	pax	300.00	3	2,700.00
Internet connection	1	lot	999.00	3	2,997.00
Sub Total					**30,697.00**
C. Training & Field work					
Training of field supervisors					
1. Lodging & Meals	2	pax	1,000.00	2	4,000.00
2. Travel costs	2	pax	100.00	2	400.00
Training of the interviewers					
1. Supervisor costs	2	pax	500.00	2	2,000.00
2. Interviewer costs	5	pax	500.00	2	5,000.00
3. Travel costs	5	pax	100.00	2	1,000.00
Field Administration					
1. Field direction	1	lot	1,500.00	2	3,000.00
2. Travel	5	pax	100.00	2	1,000.00
3.Miscellaneous, Other costs (for example, control and shipment of materials)	1	lot	2,500.00	2	5,000.00
4. Interviewers kits	1	lot	5,000.00	1	5,000.00
5. Simple token	500	pcs	50.00	1	25,000.00
Sub Total					**51,400.00**
D. Editing & Data processing					
1. Systems planning	1	lot	500.00	2	1,000.00
2. Computer programming	1	lot	1000.00	2	2,000.00
3. Clerical coding					
a. Initial coding	1	lot	500.00	2	1,000.00
b. Quality control	1	lot	500.00	2	1,000.00
c. Supervision	1	lot	500.00	2	1,000.00
4. Key to disk operations					
a. Initial keying	1	lot	500.00	2	1,000.00
b. Quality control	1	lot	500.00	2	1,000.00
c. Supervision	1	lot	500.00	2	1,000.00
5. Computer time (including operator and maintenance costs)	1	lot	1000.00	2	2,000.00
6. Miscellaneous processing costs (supplies etc.)	1	lot	2000.00	2	4,000.00
Sub Total					**15,000.00**
E. Report Writing					
1. Reproduction	1	lot	5,000.00	1	5,000.00
2. Binding	1	lot	1,000.00	1	1,000.00
3. Mailing	1	lot	1,000.00	1	1,000.00
Sub Total					**7,000.00**
F. Contingency Fund (10% of the total cost)					**22,759.70**
GRAND TOTAL					**250,356.70**

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